Transition Design Conference 2026 | Weaving Regenerative Futures | July 8-11 Mexico City | TECNOLÓGICO DE MONTERREY UNIVERSITY OF PALERMO CARNEGIE MELLON UNIVERSITY

# **Bat-Man and the Modern Smart Neighborhood:** Exploring more-than-human design and perception shifting towards Transition Design

Dinah Coops

Aalto University

dinahcoops@gmail.com

**Abstract:** Motivated by accelerating biodiversity loss, this experimental project explores whether more-than-human participatory design can help support sustainability transitions in urban systems. In particular, the experiment responds to smart neighborhood developments that prioritize technological efficiency and human convenience, while biodiversity and more-than-human needs remain peripheral. To research this, the project focused on qualitative research and evaluation through workshops in which participants *became* bats for an evening, searching for a place to live in a smart neighborhood in Kalasatama, Helsinki, Finland. The workshop aimed to assess whether inhabiting an animal's perspective can shift perceptions of what constitutes a good urban neighborhood. Findings suggested that immersive, more-than-human role-playing can alter how participants perceive and value urban space. When framed as a context-sensitive niche experiment, the workshop offers a repeatable and adaptable prototype that could inform transition planning, community engagement, and urban design education, and contribute to longer-term efforts to reimagine urban environments as mutually beneficial shared multispecies habitats.

**Keywords**: transition design; sustainability transitions; more-than-human participatory design; smart neighborhoods; urban biodiversity; multispecies urbanism

## 1. Introduction

Biodiversity is declining worldwide, and acutely so in cities (Kotze, 2023), with biologists warning that the sixth mass extinction is already underway (Barnosky et al., 2011). Although scientists and international policy organizations have repeatedly called for action (Popescu, 2015), progress toward biodiversity goals remains limited (Accountability Framework, 2024). This loss is occurring even in smart neighborhoods, which promise sustainable urban futures through the digital integration of services and infrastructure (Evans et al., 2019). In practice, however, smart neighborhood initiatives often prioritize technological optimization for human convenience, while biodiversity and more-than-human needs remain peripheral or cosmetic (Foth, 2026). Habitat fragmentation and reduced connectivity continue to threaten

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International Licence.

urban biodiversity (Kotze, 2023), even in locations that purportedly place a high value on nature, such as the state-of-the-art smart neighborhood of Kalasatama in Helsinki, Finland, where this study was originally situated. Smart urban areas such as this are important sites for research, as they often serve as models for future urban development, even when improvements are needed.

Increasingly, it is understood that strategic urban green space planning can help remedy biodiversity loss (Vallinkoski, 2023), suggesting that smart neighborhoods like Kalasatama could play a crucial role in biodiversity conservation (Soanes & Lentini, 2019; Spotswood et al., 2021). Concurrently, from a design perspective, there has been growing interest in addressing biodiversity loss and the increasing human disconnection from nature through the development of more-than-human participatory design approaches (Heitlinger et al, 2025; Metcalf, 2025). A rich opportunity exists to more actively utilize these design approaches in urban settings, both to highlight the urgency of biodiversity loss and to reframe how human urban residents, now constituting a majority of the earth's population (United Nations Statistics Division, 2018), relate to nature in their everyday environments.

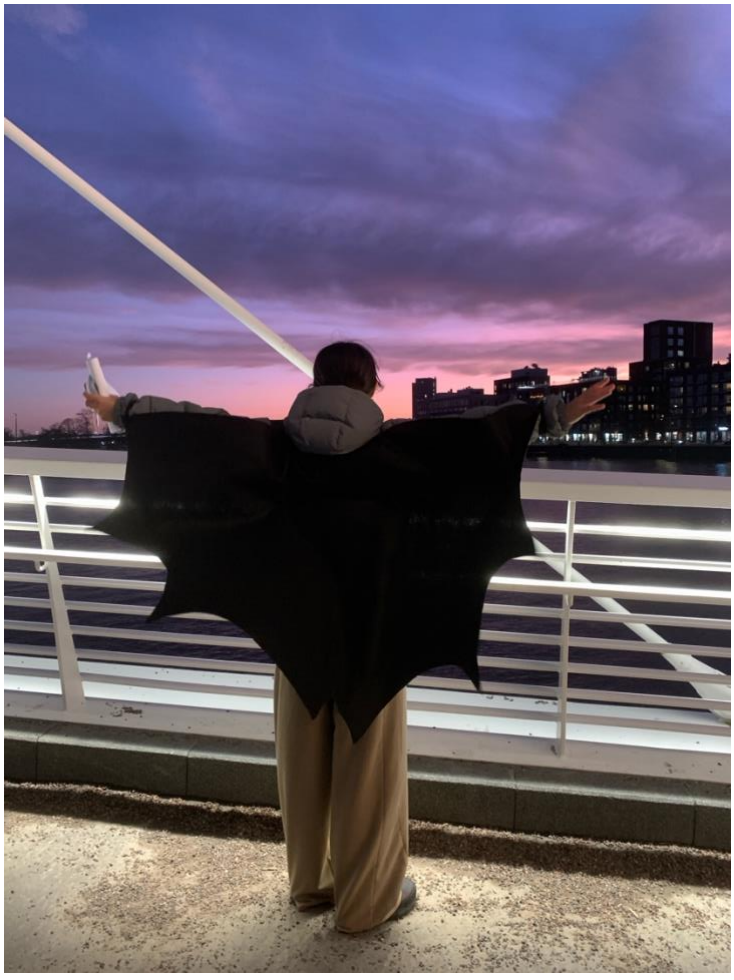


*Figure 1 Workshop participant imagining being a bat in the smart neighborhood of Kalasatama in Finland. Image by an anonymous participant.*

However, most participatory design approaches in government and institutional sustainability work remain exclusively human-centered rather than life-centered and often fail to address long-term transition horizons (Clarke et al., 2019).

Alternatively, stronger consideration of more-than-human needs has been shown to support sustainability by revealing mutual dependency and suggesting shared beneficial design solutions. Yet, there remains a need to foster real shifts in the mindsets of those responsible for smart neighborhood design (Clarke et al., 2019). Arguably, nurturing human motivation to become more in sync with our natural habitat and resources, for both resource efficiency and increased life contentment, is at least as

important as smart technology innovation in transitioning to a positive future as promised by the smart neighborhood. As a study in a Paris smart neighborhood revealed, inhabitants prioritize elements of nature and nature experiences in their urban lives over time-related conveniences, and even consider biodiversity, well-being, green corridors, and ecological continuity important (Dakouré et al., 2023).

Consequently, this experimental project proposes that more-than-human participatory design can function as a small but strategic transition design intervention to support long-term sustainability transitions in urban systems. The project asks what it would mean to consider a smart neighborhood as a shared habitat and to codesign with nonhuman residents—in this case, northern bats—as key stakeholders. By inviting participants to reimagine the Helsinki smart neighborhood of Kalasatama from the perspective of a northern bat, it explores how immersive, role-playing workshops can shift perceptions of urban spaces from exclusively human-centered needs toward multispecies, habitat-oriented futures. It lays the groundwork for how such shifts can catalyze systemic, nature-inclusive urban change and mutual benefits.

## 2. Project Relationship to Transition Design

The intent of this project directly ties to the concerns of transition design and sustainability transitions, which focus on how societies can intentionally move from unsustainable socio-technical systems toward more equitable and regenerative futures (Thomas, 2025). Transition design emphasizes long-term horizons, systemic perspectives, new ways of designing and imagining the future, and the importance of shifting underlying worldviews and values (Irwin et al, 2022). This project aligns with this point of view by considering the workshop as one element in a longer transition pathway: it operates at the level of mindsets and narratives that underpin physical configurations, and generates actionable insights that could help designers, planners, and communities reconsider urban space as shared habitat. Rather than claiming that a single workshop produces systemic change, the project positions such interventions as necessary early steps that can, over time, influence planning briefs, policy requirements, professional practices, and ultimately the design of cities.

Transitions are often understood as long-term, multi-dimensional design processes in which established socio-technical regimes—such as energy, mobility, or urban development—shift toward more sustainable design and processes resulting in system changes (Irwin, 2015; Ceschin & Gaziulusoy, 2019). These transformations often begin with niche innovations (Geels, 2002). As such, this project can be considered a niche-level experiment situated within the regime of smart urban development and the landscape of ecological crisis. It tests whether a specific design practice (more-than-human role play, supported by embodied props and reflective evaluation) can help cultivate the kinds of competencies, connections, and stories that enable broader changes in rules, norms, and infrastructures. In this sense, the workshop is not an isolated activity but a prototype for repeatable, context-sensitive

methods that could be embedded in planning processes, community engagements, or educational programs concerned with urban transitions.

## 3. Methodology

The main focus of this experiment was on designing, executing, and evaluating a workshop in which participants became northern bats navigating and inhabiting a smart neighborhood. In participatory design research, representation tools are entities or experiences that stand in for a more-than-human because they cannot represent themselves in a human-focused world (Demir & Coşkun, 2025). In this case, the representation tool of imagining oneself as a bat was chosen in part because bats are good bioindicators. By definition, good bioindicators are species whose health reflects the overall condition of the ecosystem in which they live (Parmar et al., 2016). Picking a singular bioindicator proxy species for this workshop presents a kind of shortcut, because it allows for a deeper understanding of one species while also shedding light on the surrounding ecosystem.

Background context and content for the research were gathered through a literature review and four subject matter expert interviews in Helsinki: a Chiropterologist (bat expert), a professional landscape architect, an urban planning specialist at the Ministry of the Environment, and the Project Director for Kalasatama. Each interview provided valuable background information on the interconnections between urban design and planning in Helsinki and the smart neighborhood of Kalasatama, as well as how nature was considered. The Chiropterologist was also essential for fact-checking and expanding on bat facts that were researched online regarding the northern bat's behavior and living requirements in each of the four seasons, as needed for the bat fact presentation portion of the workshop. Pre-workshop surveys were used to capture baseline perceptions of urban nature, bats, and smart neighborhoods, and to prime participants for the knowledge to be shared.

Two four-hour evening workshops were held in the smart neighborhood of Kalasatama in Helsinki, Finland. The first was with seven Creative Sustainability Aalto University students and one bat biologist, and the second, an iterative workshop from the first, was composed of two urban planning students and two urban design professionals. Several months later, two additional two-hour workshops were held. The third took place at Aalto University summer school in a Designing with Nature class in Helsinki, Finland. The fourth was at the University of Washington in Seattle, Washington, as part of a Participatory Design Class. This workshop experimented with the workshop's transferability to the United States, with students imagining themselves as the locally common little brown bat.

*Table 1 Below is an overview of the four workshops describing their location, purpose, participant count, and urban setting.*

| Location | Purpose | Participants | Urban Setting |
|---|---|---|---|
| Kalasatama, Helsinki Finland | Development and data collection | 7 creative sustainability students; 1 bat biologist | Smart neighborhood |
| Kalasatama, Helsinki Finland | Development and data collection | 4 urban planners (2 students and 2 professionals) | Smart neighborhood |
| Aalto University, Otaniemi, Helsinki, Finland | Data collection and setting testing | 14 students Designing with Nature class | Urban campus setting |
| University of Washington, Seattle, Washington, US | Data collection and transferability testing in the US | 8 students Participatory Design class | Urban campus setting |

The workshop design was composed of four main parts:

1. Slide presentation of bat facts, emphasizing the attributes and needs of bats, and giving instructions for the workshop activities.
2. Outdoor activity of exploration and searching (as a bat) for a suitable place to roost in the urban area, thinking about what it might take to make things better for a bat, and documenting the experience in the participant's choice of media.
3. Time for reflection, with options to draw or write about the experience, what they found, and if they had ideas for how to make things better for the bats.
4. Group sharing of experiences, activity results, and discussion. In the four-hour versions, feedback was also then solicited for workshop improvements.

# 4. Results

## *4.1 Overview*

The research presented here is qualitative, based on both individual participant expressions and themes shared across participants. In some cases, divergent worldviews and experiences yielded especially rich insights. Experiencing the urgency of locating a safe roosting site, combined with the acquired concrete knowledge of bats' needs (e.g., roosting conditions, light and noise sensitivity, connectivity of green spaces), appeared to encourage participants to see the neighborhood as a living habitat rather than a neutral background for

human activity. In brief, the analysis of qualitative findings from the workshops indicated that immersive, more-than-human role-playing can shift how participants perceive and value urban space.

### *4.2 Bat Perceptions*

Shifts in perceptions of the urban space appeared to begin with a shift in participants' perceptions and appreciation of bats. For example, the word clouds below document the shift in participants' perceptions of bats in the second workshop. On the left are the participants' pre-workshop survey responses to the question "What are three words that come to mind when you think of bats?" and on the right, their post-workshop responses. Here, it can be observed that the participant's pre-workshop descriptors included stereotypes and words such as "crazy," "bacteria," and "viruses," however, these disappear in the post-workshop survey. In contrast, words like "strong adaptability", "essential," and words depicting more understanding appear. Additionally, the responses from the post-workshop survey convey greater nuance, complexity, and appreciation, while still including the mystery and unknowability of the northern bats.




*Figure 2 On the left are the participants' pre-workshop survey responses to the question "What are three words that come to mind when you think of bats?" and on the right are their post-workshop responses. Word clouds generated through wordclouds.com*

Participants brought stigmas about bats to the workshop, but as they experienced and reflected on it, the idea of *living with bats* in urban spaces became a more acceptable possibility. Notably, some participants expressed limits to their tolerance of bats. These limits were expressed in terms of both the number of bats and proximity to their living space. Four participants who had been exposed to large bat populations in the past, in particular, raised concerns about the smell and potential health risks posed by waste from millions, or even just dozens, of bats, and about the hard lines that could be crossed.

Nevertheless, even these participants, after the workshop, shared that they would welcome bats living nearby; however, not too close. The workshop method appeared to dispel fears and myths about bats while helping to reintroduce creaturely sensing, understanding, and connection. As one participant, a student in Spatial planning and Transportation Engineering, commented, “Since the Bat workshop, one thought that has truly resonated with me—as a bat—is how deeply connected I am to the natural world.”

### *4.3 Visceral Learnings in the Urban Design Context*

Embodied, visceral experiences such as this also appear to foster a deeper, different level of learning and create a strong argument for using more-than-human participatory design as a means to achieve it. Another participant had wondered why urban planners and designers do not simply rely on expert consultation and a set of criteria to accommodate more-than-human needs in their plans. However, reflecting on the workshop, they wrote, “I have realized that it is necessary to understand the needs of other species on a more-than-factual basis. And especially after today, I internalized that a lived experience far outweighs fact on paper.”

The first research site in the smart neighborhood of Kalasatama, Helsinki, Finland, was decidedly not built with bats or any more-than-humans in mind. The challenge of searching for bat homes in this neighborhood, and observing that very little would work well for them, could have led to complete frustration for participants. However, the lack of green corridors and nature necessary for survival as a bat appeared to force creative and thoughtful solutions, and created empathy and concern while role-playing as these creatures attempting to find a quiet and safe place to sleep.

Indeed, a majority of participants described a new sense of positive symbiosis between bats, humans, and urban environments, and several reported increased motivations to advocate for more ecologically mindful urban design. Additionally, they made the leap of thought that considerations for more-than-humans should be taken into account when planning and designing urban spaces. As one participant expressed in the post-workshop survey, “I think that there can be a symbiosis of some species, like bats, and urban spaces…we just need to take them into account when planning and designing those areas.” Another participant wrote, “When I explored our city as a bat, I found that bats’ living requirements don’t have a lot of conflict with humans”. They noted that bats “are active at night” when humans are not and that they can adapt “to live in urban constructions”. Additionally, they expressed that we, as humans, “can take a more detailed consideration when we build these artificial environments to make them suitable for non-humans to live as well.”

The workshop also raised awareness that current smart neighborhood visions and infrastructure are not designed from a biodiversity perspective, highlighting contradictions between human-centered technological smartness and ecological resilience. One participant summed up this perception change in this way: “I think I'm going to be looking at a lot of

urban areas differently now, and I just feel a lot of sorrow that an area that's kind of praised for its being environmentally forward doesn't really have any impactful green areas."

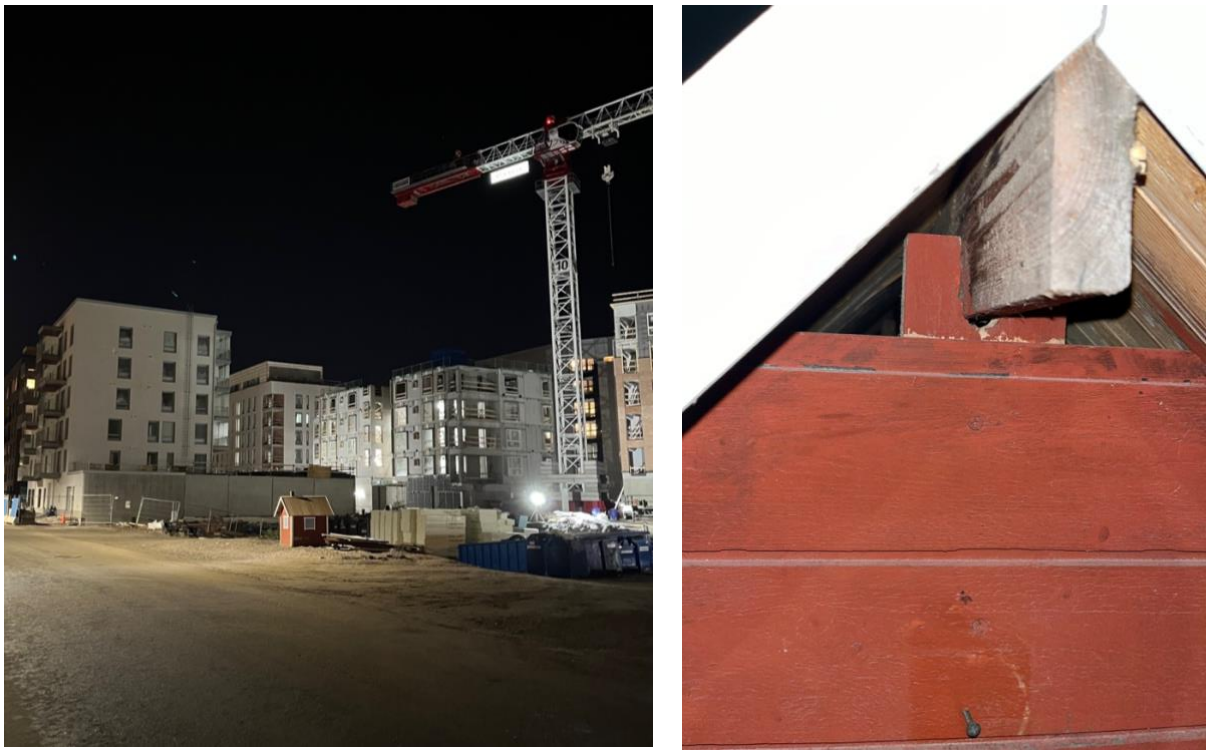


*Figure 3 A tiny house designed for the marketing of a new co-housing construction as a solution for a maternity colony for 24 northern bats and pups, due to its size and bat-sized openings in the eaves. The participant also shared in their notes, "I would like bats in my building. I change my mind. I want them to be safe." Image credits: Miina Heikkinen, with permission.*

Important to note, this workshop was originally designed to be integrated into urban planning student curricula. The ideal participants sought were urban planning students, as they were believed to have the greatest potential impact over a long-term horizon. However, after conducting the study in four variations in two countries with a variety of participants, it was observed that reconnections and appreciation of nature occurred to some degree with all participants. This appeared to demonstrate that the method also has the potential to create community-driven pressure on the design of urban neighborhoods.

## 5. Limitations

Understandably, there are anthropomorphic limits to what humans can imagine, relate to, and tolerate in a workshop such as this. In the philosophical and highly influential paper "What Is It Like to Be a Bat?", Nagel (1974) outlined the well-formed argument that it is impossible for humans to experience the world from the perspective of other species. We cannot echolocate, capture a fly in mid-air, or experience a mating swarm. However, more-than-human designers have believed that, while embracing this unknowability, it is still valuable to try (Dunne & Raby, 2024). Even if none of what we imagine through the lens of

being human in trying to understand a bat's senses and needs are accurate, the act of trying, based on our limited knowledge, appeared in these workshops to create a sense of empathy and *being with* the bats.

Even though this act of trying to imagine what a bat needs and experiences was the critical part of a workshop like this, there are limitations to what we as humans can tolerate. In these workshops, some participants experienced fear when exploring dark and unknown spaces and had concerns about physical threats or injury. Interestingly, participants who spent most of their time exploring on their own had the strongest sense of imagining themselves as a bat but were also more likely to feel fear as a human. On a practical level, in later workshops, it was suggested to spend at least some of the workshop time on one's own and to explore only within the bounds of one's personal sense of safety.

Another important limitation to report is that, while the outcomes of the workshop appeared surprisingly effective in a short timeframe, this study did not evaluate whether these changes in mindset persist over time.

## 6. Methodological Findings

Through the original prototype workshop and revised second workshop, practical findings emerged, and improvements were made along the way. Ideas and suggestions from participants for improving the workshop were solicited both during the final stage of the four-hour workshops and in the post-workshop survey. A few examples are as follows.

Bat wings had not yet been made for the first workshop, and participants suggested adding more props to help participants get into character as an improvement. The delight in the second workshop, with participants wearing the wings and commenting on how it helped them connect with bats, provided evidence that this made a significant difference. Also, in the first workshop, it was suggested that having more art supplies invitingly displayed and available for reflective documentation after the outdoor exploration would be desirable. However, for the second workshop, the participants found their presence intimidating. During the feedback session of this workshop, one participant suggested asking participants to produce a one-minute drawing, as it takes performance pressure off. Trying this immediately, even the most skeptical participant described this, surprisingly, as the most poignant part of the workshop for them. Documenting in general appeared to concretize the experience for participants. Additional suggestions from the first workshop included providing a more detailed handout for the outdoor exploration. One suggestion was to include exactly what the insects the northern bat hunts look like. Another was to show and describe exactly what size opening a northern bat needs to enter to find a safe roosting site. Having very specific, tangible examples of what to look for or imagine helped the exercise feel more viscerally real for participants.

Another limitation learned in the methodology was that, to be effective, more-than-human participatory workshops are time-consuming for participants, which can be a barrier to

recruitment and scheduling. The first two workshops were four hours in duration, providing adequate time for information sharing, exploration, reflection, and discussion. The subsequent two workshops were designed to be two hours to fit into academic schedules. While these shorter workshops elicited some changes in perception, the longer workshops allowed for more reflection and deeper shifts.



*Figure 4 Sharing the experience and ideas from the outdoor activity of finding a roosting place in Kalasatama with the group. Image credit: Steve Havas, with permission.*

# 7. Discussion/Implications

## *7.1 Contributions to Transition Design Discourse*

This project contributes to transition design discourse in at least three ways. First, it offers an empirically grounded example with methodological contributions of how more-than-human participatory methods can be designed and evaluated as transition interventions, rather than treated only as speculative or artistic exercises. Using pre- and post-surveys and qualitative analysis, the research explicitly assesses whether and how the intervention changes perceptions and motivations. Second, the methodology describes practical design tools and techniques—priming surveys, species-specific knowledge scaffolding, embodied props, open-ended self-documentation—that can be recombined and adapted in other projects aiming to reimagine human–nature relations over time. Third, in the original format, it situates these tools within the place-based smart neighborhood context of Kalasatama in Helsinki, demonstrating how local species, policies, and planning shape both the challenges and possibilities of transition design.

## *7.2 Systems Change Lens*

From the perspective of what systems change means here, the project primarily intervenes in non-material aspects of the urban system—perceptions, imaginaries, and values—but does so with intention towards physical follow-through. By reorienting participants to consider nocturnal flight paths, roosting niches, safe corridors, and the effects of lighting and

sound from a bat's perspective, the workshop generates concrete design implications: for example, adjustments to façade or roof design to provide roosting sites, revised lighting strategies that support bats' food supply and sensitivities, or the integration of vegetated corridors that support insect populations and bat movement. While these suggestions are prompted by the needs of bats as identified by the bat expert, implementation ideas and constraints arose from the participants. The project shows how more-than-human participatory workshops can produce design concepts and criteria that could be translated into spatial guidelines, technical standards, or policy recommendations when adopted by municipalities, developers, or housing associations.

The project also engages with issues of social relations and power, which are central to transition design. In practice, urban planning decisions are shaped by inequalities between institutional actors (planners, developers, technology companies), residents, and nonhuman life, which has no direct voice. By explicitly naming bats as codesigners and centering their needs in the design exercise, the workshop changes the default hierarchy of interests and invites participants to reflect on whose needs are typically prioritized. The process opens up questions about who can influence building codes, who benefits from smart technologies, and who bears the costs of ecological degradation. Here, tentative principles emerge: use playful, embodied methods to simplify complex ecological knowledge and include more-than-human, less powerful, and absent stakeholders in systems mapping and design.

### *7.3 Place, Culture, and Transferability*

Place and culture are central to this approach. For example, Kalasatama's smart neighborhood context is characterized by strong municipal planning traditions, a technological innovation agenda, and particular climatic and ecological conditions (including the presence of protected bat species). The choice of the northern bat as the focal species reflects local conservation concerns and makes the workshop immediately relevant to participants' everyday environment. At the same time, the method is deliberately designed for transferability: other cities and communities can substitute species that are ecologically and culturally significant in their own contexts—such as swifts, pollinators, or urban mammals—and reframe the workshop around local planning processes and smart technologies.

## 8. Future Evolutions

Looking ahead, there are several ways the project could be reimagined and expanded through a transition design lens. Iterative cycles of the workshop could be conducted with different actor groups—city planners, developers, community organizations, and schools—to understand different perspectives and where alliances might form. Notably, transferring the workshop to high-level stakeholders would require research, testing, and adjustments, as well as some resource investment to adapt it to a different audience. Ideally, when engaging decision makers, the workshop would be part of a greater effort with a future

vision that encompasses long-term benefits. The method could ultimately be integrated into smart neighborhood planning as a required step in concept development or public engagement, ensuring that multispecies perspectives inform key decisions rather than being added after the fact. Partnerships with citizen science initiatives or biodiversity monitoring programs could connect perceptual shifts with ongoing data collection and stewardship, deepening participants' engagement beyond the workshop. Over time, such interventions could contribute to a broader culture of nature-inclusive urban planning, supporting a transition to ecologically grounded decision-making in cities and neighborhoods.

## 9. Conclusion

This experimental project invited workshop participants to role-play living in a smart neighborhood from a bat's perspective and to imagine how it could be different. It explored how immersive, role-playing workshops can shift perceptions of urban spaces from exclusively human-centered needs toward multispecies, habitat-oriented futures, and how such shifts can lay the groundwork for systemic, nature-inclusive urban change and mutual benefits. It takes the perspective that nurturing human motivation to live in sync with biodiverse natural habitats is as important as technological innovation in smart neighborhoods for enabling sustainable living. Given this, the project points toward narratives and strategies that could help persuade funders, sponsors, and institutions to support transition-oriented work with longer time horizons. It demonstrates that relatively modest, well-designed interventions can generate measurable shifts in perception and motivation regarding biodiversity and make visible design directions with potential co-benefits for climate resilience and public health. Positioning such work as a low-cost, high-leverage complement to infrastructure investments, and emphasizing its evaluative rigor and imaginative power, can make a compelling case for sustained support. In this way, the project not only explores what it means to design with and for bats in a smart neighborhood, but also contributes to broader conversations about how design can help societies navigate and shape long-term sustainability transitions.

Acknowledgments: I would like to express my gratitude to my thesis supervisor, İdil Gaziulusoy, and my advisors Emīlija Veselova and Andrea Botero for their guidance throughout the development of my thesis, culminating in this research paper. Additionally, I would like to thank my insightful workshop participants for their enthusiastic participation and my fellow Aalto University Creative Sustainability classmates for their encouragement and critical feedback. Finally, I thank my husband, Steve Havas, for his unwavering support in this journey with the bats.

## 10. References

Accountability Framework. (2024, November 7). *COP16: More progress needed on global biodiversity efforts to meet 2030 goals*. Retrieved June 5, 2026, from https://accountability-framework.org/pt/news-events/news/article/cop16-more-progress-needed-on-global-biodiversity-efforts-to-meet-2030-goals/

Barnosky, A. D., Matzke, N., Tomiya, S., Wogan, G. O., Swartz, B., Quental, T. B., ... & Ferrer, E. A. (2011). Has the Earth's sixth mass extinction already arrived?. *Nature*, *471*(7336), 51–57.

Ceschin, F., & Gaziulusoy, İ. (2019). *Design for sustainability: a multi-level framework from products to socio-technical systems*. Routledge.

Clarke, R., Heitlinger, S., Light, A., Forlano, L., Foth, M., & DiSalvo, C. (2019). More-than-human participation. *Interactions*, 26(3), 60–63. https://doi.org/10.1145/3319075

Dakouré, A., Bourdeau-Lepage, L., & Georges, J. (2023). The Paris urban plan review: an opportunity to put the 15-Minute City concept into the perspective of the Parisians' desire for nature, in *Elsevier eBooks* (pp. 61–75).

Demir, B. S., & Coşkun, A. (2025). Rethinking representation in design: Towards constructing parameters for representation tools in more-than-human design. In *Proceedings of the Designing Interactive Systems Conference (DIS '25)*, July 05–09, 2025, Funchal, Portugal (pp. 1–17). ACM. https://doi.org/10.1145/3715336.3735680

Dunne, A. & Raby, F. (2024, November 29). *Thomas Nagel revisited*. Future Observatory Journal. https://fojournal.org/re-reading/thomas-nagel-dunne-and-raby/

Evans, J., Karvonen, A., Luque-Ayala, A., Martin, C., McCormick, K., Raven, R., & Palgan, Y. V. (2019). Smart and sustainable cities? Pipedreams, practicalities and possibilities. *Local Environment*, 24(7), 557–564.

Foth, M. (2026). A scientist's warning on smart cities: rethinking urban sustainability for more-than-human futures. *Journal of Urban Technology*, *33*(1), 141–157.

Geels, F. W. (2002). Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. *Research policy,* 31(8-9), 1257–1274.

Heitlinger, S., Light, A., Akama, Y., Lindström, K., & Ståhl, Å. (2025). More-than-human participatory design. *Routledge International Handbook of Contemporary Participatory Design*, 79–110.

Irwin, T. (2015). Transition design: A proposal for a new area of design practice, study, and research. *Design and culture*, 7(2), 229–246.

Irwin, T., Tonkinwise, C., & Kossoff, G. (2022). Transition design: An educational framework for advancing the study and design of sustainable transitions. *Cuadernos del Centro de Estudios en Diseño y Comunicación. Ensayos*, (105), 31–72.

Nagel, T. (1974). What Is It Like to Be a Bat? *The Philosophical Review*, 83(4), 435–450. https://doi.org/10.2307/2183914

Parmar, T. K., Rawtani, D., & Agrawal, Y. K. (2016). Bioindicators: the natural indicator of environmental pollution. *Frontiers in Life Science, 9*(2), 110–118. https://doi.org/10.1080/21553769.2016.1162753

Popescu, O. C. (2015). United Nations decade on biodiversity: strategies, targets and action plans. *Urbanism. Arhitectură. Construcţii*, *6*(2), 37–50.

Soanes, K., & Lentini, P. E. (2019). When cities are the last chance for saving species. *Frontiers in Ecology and the Environment*, *17*(4), 225–231.

Spotswood, E. N., Beller, E. E., Grossinger, R., Grenier, J. L., Heller, N. E., & Aronson, M. F. (2021). The biological deserts fallacy: cities in their landscapes contribute more than we think to regional biodiversity. *BioScience*, *71*(2), 148–160.

Thomas, S. (2025). Futures thinking and regenerative sustainability. In *The Routledge Handbook of Global Sustainability Education and Thinking for the 21st Century* (pp. 307–324). Routledge India.

United Nations. (2018). *68% of the world population projected to live in urban areas by 2050, says UN | United Nations.* https://www.un.org/en/desa/68-world-population-projected-live-urban-areas-2050-says-un

Vallinkoski, A. (2023, September 25). Urban nature is surprisingly diverse. *University of Helsinki*. https://www.helsinki.fi/en/news/biodiversity-loss/urban-nature-surprisingly-diverse

## About the Author:

**Dinah Coops** is a lifelong experimenter coming from a background in graphic design, art direction, sustainable product design, and human-centered design. After completing a Creative Sustainability master's degree at Aalto University, she's currently exploring more-than-human urban applications in Seattle, Washington.